From Novel View Synthesis to Generative World Modeling: A Comprehensive Survey

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Abstract

Novel View Synthesis (NVS)—rendering a scene from arbitrary unseen viewpoints given reference observations—has undergone a paradigm shift from classical geometry-based reconstruction toward learning-driven generative approaches. This survey examines NVS not merely as a rendering technique but as the **foundational spatial capability** of Generative World Models: systems that predict visual scene states under arbitrary conditions including viewpoint changes, temporal dynamics, and agent interactions.

We present a unified conceptual framework organising existing methods along a *continuous spectrum*—from static multi-view reconstruction to dynamic, interactive world generation—across three axes: (i) scene representation (implicit, explicit, hybrid), (ii) generative paradigm (per-scene optimisation, feed-forward inference, diffusion-based synthesis), and (iii) world-modelling capability (geometry, dynamics, controllability). We survey more than **160 representative methods** spanning NeRF, 3D Gaussian Splatting, generalizable reconstruction, diffusion-based NVS, and 4-D world modeling, identify the key open challenges, and propose a roadmap toward physically grounded, generalisable world models.

1 Introduction

1.1 The Fundamental Problem

Intelligence requires reasoning about the world beyond what is immediately visible. A self-driving vehicle must anticipate the scene around a blind corner; a robot must predict how a grasped object will appear from the wrist camera; a surgeon must mentally rotate an anatomy from a 2-D scan. In each case the core operation is identical: **synthesising a plausible visual state at an unobserved vantage point**.

This is the problem of **Novel View Synthesis (NVS)**. Given reference images $\{\mathbf{I}_i, \boldsymbol{\pi}_i\}_{i=1}^n$ with known camera poses, the goal is to render a photo-realistic image $\hat{\mathbf{I}}$ at an arbitrary target pose $\boldsymbol{\pi}^*$, faithfully preserving geometry, appearance, and lighting.

While the problem was studied in the image-based rendering era [1–4], the past five years have witnessed an extraordinary leap: NeRF [17], 3DGS [46], and diffusion-based methods [91, 102] have pushed the frontier from slow per-scene reconstruction toward fast, generalisable, and generative synthesis.

Preprint.

Table 1: Comparison with related surveys. ✓ covered; ∼ partially; – not covered.

Survey	Year	NeRF	3DGS	Generative	Dynamic	Generalise	4D/WM	Unified
Tewari *et al.* [9]	2022	✓	–	∼	∼	∼	–	–
Gao *et al.* [10]	2022	✓	–	–	∼	–	–	–
Zhu *et al.* [11]	2023	∼	–	–	–	–	–	–
Wu *et al.* [12]	2024	–	✓	–	∼	–	–	–
Fei *et al.* [13]	2024	–	✓	∼	∼	–	–	–
Chen *et al.* [14]	2024	∼	∼	–	✓	–	∼	–
Zhang *et al.* [15]	2025	∼	∼	–	–	✓	–	–
He *et al.* [16]	2025	–	✓	∼	∼	–	–	–
Ours	2026	✓	✓	✓	✓	✓	✓	✓

1.2 From NVS to World Modeling

We argue that this progression reflects a deeper conceptual shift: **NVS is becoming the spatial inference engine of Generative World Modeling**.

> **Working Definition (World Model).** A *Generative World Model* is a learned system that can predict the visual (and optionally physical) state of a scene under arbitrary conditions—viewpoint changes, temporal evolution, and agent interactions—in a manner that is *geometrically consistent*, *physically plausible*, and *generalisable* beyond the training distribution.

Under this definition, NVS is the special case where the only varying condition is camera pose. A world model must additionally handle temporal dynamics and action conditioning. Crucially, *spatial consistency enforced by NVS is a prerequisite*: a world model without geometric grounding is merely a video generator—plausible in appearance but unreliable for spatial reasoning.

1.3 Relationship to Prior Surveys

Table 1 compares this work with the most closely related prior surveys. Tewari *et al.* [9] and Gao *et al.* [10] focus on NeRF representations without addressing world modelling. Wu *et al.* [12] and Fei *et al.* [13] cover 3DGS but not generative methods or 4-D. Chen *et al.* [14] cover NVS for autonomous driving but lack a world-modeling lens. To our knowledge, **no existing survey jointly covers all three paradigms (NeRF, 3DGS, generative) through the unifying lens of Generative World Modeling**, nor proposes a continuous-spectrum taxonomy from static reconstruction to 4-D interactive generation.

1.4 The Unified Spectrum

We organise all methods along a *continuous spectrum* defined by three axes (Figure 1):

Axis 1. Input regime: dense multi-view → sparse-view → single image → text/unconditional.

Axis 2. Temporal scope: static scene → deformable/dynamic → 4-D world generation.

Axis 3. Controllability: viewpoint only → viewpoint + time → viewpoint + time + action.

1.5 Contributions and Paper Structure

Contributions: (1) Unified continuous-spectrum taxonomy with three axes; (2) Cross-paradigm coverage of >160 methods (NeRF, 3DGS, generative, 4-D world modeling); (3) Explicit related-survey gap analysis (Table 1); (4) Open-problems roadmap with five concrete research directions.

Structure: §2 Background; §3 Scene Representations; §4 NeRF Methods; §5 3DGS Methods; §6 Generalisable Feed-Forward Reconstruction; §7 Generative NVS; §8 Towards World Modeling; §9 Datasets; §10 Evaluation; §11 Open Problems; §12 Conclusion.

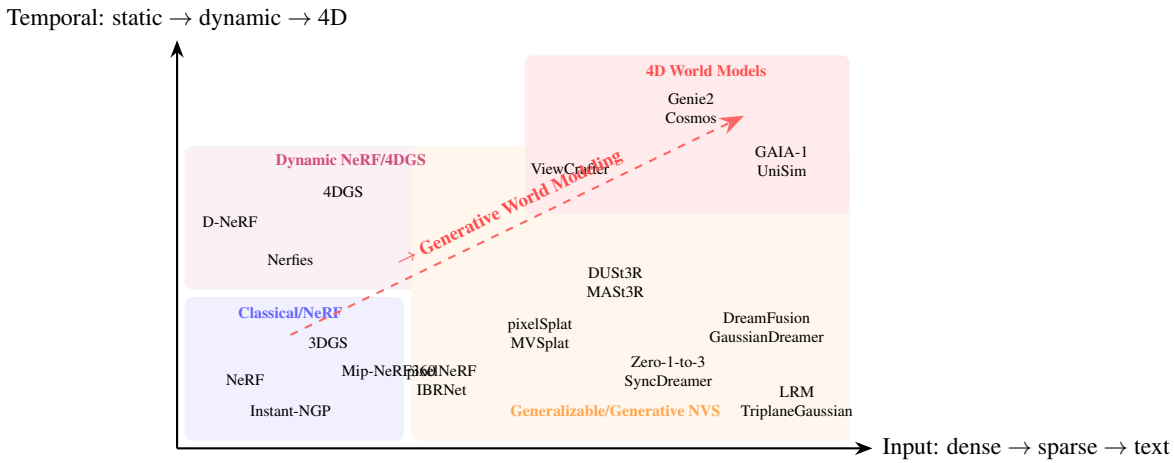

Figure 1: The unified NVS spectrum. The diagonal arrow marks the evolutionary trajectory toward 4-D Generative World Modeling.

2 Background and Unified Formulation

2.1 Classical Image-Based Rendering

Before the neural era, novel view synthesis was approached through *image-based rendering* (IBR) [1–3]. Light Field Rendering [1] densely samples the plenoptic function [6] to enable view interpolation without explicit geometry. View-dependent texture mapping [4] and Lumigraph [5] extended these ideas. Structure from Motion (SfM) [7] and Multi-View Stereo (MVS) [8] provided the geometry backbone. The key limitation of classical IBR is its dependence on dense capture: generalisation to novel poses requires interpolation within the observed set, not extrapolation beyond it.

2.2 Volume Rendering and NeRF

NeRF [17] represents a scene as a continuous function $F_\theta : (\mathbf{x}, \mathbf{d}) \mapsto (\mathbf{c}, \sigma)$. The expected colour of a ray $\mathbf{r}(s) = \mathbf{o} + s\mathbf{d}$ is:

$$\hat{C}(\mathbf{r}) = \int_{s_n}^{s_f} T(s)\,\sigma(\mathbf{r}(s))\,\mathbf{c}(\mathbf{r}(s), \mathbf{d})\,\mathrm{d}s, \quad T(s) = \exp\left(-\int_{s_n}^{s} \sigma(\mathbf{r}(u))\,\mathrm{d}u\right). \tag{1}$$

The training objective minimises photometric reconstruction loss. Stratified and hierarchical sampling make the continuous integral tractable.

2.3 Gaussian Splatting

3DGS [46] represents a scene as N anisotropic Gaussians $\{\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i, \alpha_i, \mathbf{c}_i\}$. Pixel colour is computed by front-to-back alpha compositing of projected Gaussians:

$$\hat{C} = \sum_{i \in \mathcal{N}} \mathbf{c}_i\, \alpha_i\, \tilde{G}_i \prod_{j<i} (1 - \alpha_j\, \tilde{G}_j), \tag{2}$$

where $\tilde{G}_i$ is the 2-D projection of Gaussian i. Adaptive densification clones/splits under-reconstructed regions and prunes redundant Gaussians.

2.4 Unified NVS and World Modeling Formulation

We formulate NVS as a general function:

$$\mathcal{M}_\theta : \left(\{\mathbf{I}_i, \boldsymbol{\pi}_i\}_{i=1}^{n},\ \boldsymbol{\pi}^*\right) \longrightarrow \hat{\mathbf{I}}^*, \quad n \in [1, +\infty). \tag{3}$$

The world-modeling extension adds time t and action $\mathbf{a}$:

$$\mathcal{M}_\theta : \left(\{\mathbf{I}_i, \boldsymbol{\pi}_i\}_{i=1}^{n},\ \mathbf{a}_{0:t},\ t,\ \boldsymbol{\pi}^*\right) \longrightarrow \hat{\mathbf{I}}_t^*. \tag{4}$$

This defines five desiderata: **D1** geometric consistency; **D2** sparse/single-image generalisation; **D3** dynamic scene modelling; **D4** action conditioning; **D5** physical plausibility.

3 Scene Representation Taxonomy

NVS methods can be classified by how they represent scene content:

Implicit representations encode scenes as continuous functions (*e.g.*, MLP, hash-grid). Advantages: compact, resolution-independent, continuous. Disadvantages: slow rendering, per-scene optimisation.

Explicit representations store scene content at discrete elements (*e.g.*, voxels, point clouds, Gaussians). Advantages: fast rendering, easy editing. Disadvantages: memory-intensive, discrete resolution limits.

Hybrid representations combine implicit and explicit components (*e.g.*, triplane + MLP, Gaussian + neural field). Advantages: balance speed and quality. Disadvantages: complex training pipelines.

Structured/compositional representations decompose scenes into semantic entities (*e.g.*, objects, backgrounds, agents). Advantages: scene editing, compositional generation. Disadvantages: require segmentation or part annotation.

4 NeRF: Implicit Neural Radiance Fields

4.1 Foundational NeRF Methods

The original NeRF [17] demonstrated that an MLP with positional encoding could represent complex real scenes with view-dependent appearance. Subsequent work addressed NeRF's two principal limitations: *slow optimisation* and *poor generalisation*.

Reducing optimisation time. Instant-NGP [23] replaces the MLP backbone with a multi-resolution hash grid, reducing training from hours to seconds while achieving state-of-the-art quality. TensoRF [24] decomposes the radiance field into compact tensor components (CP and VM decomposition). Plenoxels [25] uses a sparse 3-D voxel grid with spherical harmonics, achieving minute-scale optimisation without neural networks. DirectVoxGO [26] adopts a two-stage coarse-to-fine voxel grid. DVGO [26] and TensoRF set new speed records for per-scene NeRF.

Improving rendering quality. Mip-NeRF [19] casts conical frustums instead of rays, reducing aliasing at multiple scales. Mip-NeRF 360 [20] extends this to unbounded 360° scenes via a non-linear scene contraction. Zip-NeRF [21] combines the hash grid of Instant-NGP with the anti-aliasing of Mip-NeRF 360. NeRF-W [18] models per-image appearance variation for in-the-wild captures. Ref-NeRF [22] improves specular appearance with a physically-based reflection model.

Surface and geometry extraction. NeuS [27] reformulates volume rendering using a signed distance function (SDF), yielding better surface reconstruction. VolSDF [28] provides an alternative SDF-to-density conversion. NeRFMeshing [**?**] and BakedSDF [29] extract high-quality meshes from NeRF.

Table 2 summarises representative NeRF variants.

4.2 Generalizable NeRF

Per-scene NeRF requires lengthy per-scene optimisation, limiting scalability. *Generalisable* methods learn a shared prior across scenes, enabling fast inference on new scenes.

pixelNeRF [30] conditions the NeRF MLP on pixel-aligned CNN features extracted from reference views, enabling single-image reconstruction. IBRNet [31] uses a transformer to aggregate features from multiple reference views at query points along the ray. MVSNeRF [32] builds a cost volume from stereo for geometry-aware feature extraction. GNT [33] is a fully attention-based generalizable NeRF. SRT [34] uses a set-latent scene representation encoded from arbitrary views. NerfDiff [35] augments generalizable NeRF with diffusion priors for single-image synthesis. ZeroRF [36] achieves sparse-

Table 2: NeRF variants: representative methods and properties.

Method	Venue	Train	Render	Generalise	Key Contribution
NeRF [17]	ECCV'20	Slow	Slow	–	MLP + positional enc.
NeRF-W [18]	CVPR'21	Slow	Slow	–	In-the-wild appearance
Mip-NeRF [19]	ICCV'21	Slow	Slow	–	Anti-aliased cones
Plenoxels [25]	CVPR'22	Fast	Fast	–	No-MLP voxel grid
TensoRF [24]	ECCV'22	Med	Fast	–	Tensor decomp.
DirectVoxGO [26]	CVPR'22	Fast	Fast	–	Two-stage voxel
Instant-NGP [23]	TOG'22	Fast	Fast	–	Hash grid
Mip-NeRF360 [20]	CVPR'22	Slow	Slow	–	Unbounded scenes
Ref-NeRF [22]	CVPR'22	Slow	Slow	–	Specular reflection
NeuS [27]	NeurIPS'21	Slow	Slow	–	SDF-based surface
VolSDF [28]	NeurIPS'21	Slow	Slow	–	SDF-density bridge
Zip-NeRF [21]	ICCV'23	Fast	Fast	–	Hash + anti-alias
BakedSDF [29]	SIGGRAPH'23	Med	Fast	–	Mesh extraction

Table 3: Dynamic NeRF methods.

Method	Venue	Multi-view	Monocular	Key Idea
D-NeRF [37]	CVPR'21	✓	–	Deformation network
NSFF [38]	CVPR'21	✓	∼	Scene flow field
Nerfies [39]	ICCV'21	–	✓	Latent deformation
HyperNeRF [40]	TOG'21	–	✓	Ambient dimension
DyNeRF [41]	CVPR'22	✓	–	Multi-view video NeRF
TiNeuVox [42]	SIGGRAPH'22	✓	–	Voxel dynamic accel.
K-Planes [43]	CVPR'23	✓	∼	4-D plane decomp.
HexPlane [44]	CVPR'23	✓	∼	Hexagonal planes
NeRFPlayer [45]	TOG'23	✓	–	Region decomp.

view 360° reconstruction via per-scene optimisation with zero pretraining, leveraging factorised feature grids.

4.3 Dynamic NeRF

Modelling non-rigid dynamic scenes is a central challenge for world modeling.

D-NeRF [37] introduces a deformation network mapping each spacetime point to a canonical static space. Nerfies [39] and HyperNeRF [40] model complex non-rigid deformations via latent codes and higher-dimensional ambient spaces. Neural Scene Flow Fields (NSFF) [38] models dynamic scenes with scene flow representation. DyNeRF [41] extends NeRF to monocular video via a time-conditioned model. TiNeuVox [42] uses voxel features to accelerate dynamic NeRF. K-Planes [43] decomposes 4-D space-time into six feature planes (three spatial, three spatial-temporal). HexPlane [44] is a concurrent work with the same idea. NeRFPlayer [45] separately encodes static, deforming and newly appearing regions.

Table 3 summarises dynamic NeRF methods.

5 3D Gaussian Splatting

5.1 Foundational 3DGS

3DGS [46] achieved real-time (>100 fps at 1080p) high-quality NVS with fast training. Its explicit, editable representation opened entirely new research directions.

Quality improvements. Mip-Splatting [47] applies 3-D smoothing filters to address aliasing at different viewing distances. AbsGS [48] improves densification with an absolute gradient criterion, reducing needle artifacts. Taming 3DGS [**?**] provides principled primitives for bounded and compact reconstruction.

Compression and efficiency. LightGaussian [50] globally prunes and vector-quantises Gaussians, achieving $15\times$ compression with minimal quality loss. Compact3D [51] uses k-means codebooks for attribute compression. EAGLES [52] compresses Gaussians using residual vector quantization.

Table 4: 3DGS variants.

Method	Venue	RT	Dyn.	Surf.	Comp.	Key Idea
3DGS [46]	TOG'23	✓	–	–	–	Gaussian splatting
Mip-Splatting [47]	CVPR'24	✓	–	–	–	Anti-aliased splats
AbsGS [48]	NeurIPS'24	✓	–	–	–	Absolute grad. dens.
Scaffold-GS [49]	CVPR'24	✓	–	–	✓	Anchor structure
LightGaussian [50]	NeurIPS'23	✓	–	–	✓	Pruning + VQ
Compact3D [51]	arXiv'24	✓	–	–	✓	k-means codebook
EAGLES [52]	ECCV'24	✓	–	–	✓	Residual VQ
HAC [53]	ECCV'24	✓	–	–	✓	Hash context model
SuGaR [54]	CVPR'24	✓	–	✓	–	Joint mesh+GS
2DGS [55]	SIGGRAPH'24	✓	–	✓	–	2-D disk Gaussians
GOF [56]	arXiv'24	✓	–	✓	–	Opacity field
GaussianEditor [57]	CVPR'24	✓	–	–	–	LLM-guided editing
Gaussian Grouping [58]	ECCV'24	✓	–	–	–	Identity field
GALA3D [59]	arXiv'24	✓	–	–	–	Compositional gen.

Table 5: Feed-forward generalizable Gaussian Splatting methods.

Method	Venue	Input views	Unposed	Key Idea
pixelSplat [66]	CVPR'24	2	–	Stereo epipolar feat.
MVSplat [67]	ECCV'24	2–6	–	Cost volume prior
TranSplat [68]	arXiv'24	2+	–	Transformer MV
FSGS [69]	CVPR'24	3	–	Gaussian unpooling
DepthSplat [70]	arXiv'24	2+	–	Joint depth+GS
Splatt3R [71]	arXiv'24	2	✓	DUSt3R backbone
TriplaneGaussian [72]	CVPR'24	1	–	Triplane+Gaussian

Scaffold-GS [49] introduces anchor Gaussians that drive neural Gaussian attribute prediction, substantially reducing storage. HAC [53] uses hash-grid-aided context models for extreme compression.

Surface reconstruction. SuGaR [54] jointly optimises mesh and Gaussians for high-quality surface extraction from 3DGS. 2DGS [55] uses 2-D disk Gaussians for better surface alignment. Gaussian Opacity Fields (GOF) [56] combines Gaussian Splatting with a continuous opacity field.

Scene editing and compositional scenes. GaussianEditor [57] performs language-guided 3-D Gaussian editing. Gaussian Grouping [58] segments 3DGS with identity fields for part-level editing. GALA3D [59] generates multi-object compositional scenes with 3DGS via LLM-guided layout.

Table 4 summarises key 3DGS variants.

5.2 Dynamic 3DGS

4DGS [60] extends Gaussians into 4-D space-time with a temporal radiance field. SpacetimeGaussians [61] assigns polynomial time trajectories to each Gaussian. Deformable3DGS [62] applies a deformation network to canonical Gaussians. SC-GS [63] uses sparse control points for monocular dynamic scenes. Gaussian-Flow [64] models Gaussian dynamics via polynomial motion fields in Fourier space. Shape-of-Motion [65] reconstructs per-Gaussian 3-D motion from monocular videos.

5.3 Feed-Forward Generalizable 3DGS

pixelSplat [66] predicts Gaussian primitives from stereo image pairs in a single forward pass. MVSplat [67] adds cost-volume priors for better geometry. TranSplat [68] introduces a transformer architecture for multi-view Gaussian prediction. Splatt3R [71] builds on DUSt3R [73] for unposed image pairs. FSGS [69] enables few-shot Gaussian Splatting via Gaussian unpooling and monocular depth supervision. DepthSplat [70] jointly predicts depth and Gaussians for generalizable NVS. TriplaneGaussian [72] combines triplane and Gaussian representations in a transformer-based single-view reconstruction model.

Table 5 summarises feed-forward generalizable 3DGS.

6 Generalisable Feed-Forward 3D Reconstruction

6.1 Pose-Free and Foundation Models

A new generation of methods aims for *pose-free* generalizable reconstruction. DUSt3R [73] predicts dense pointmaps from arbitrary image pairs without camera calibration, then aligns them via global optimisation. MASt3R [74] adds matching capabilities to DUSt3R, enabling accurate localisation. MonST3R [75] extends DUSt3R to dynamic video. VGGT [76] is a visual geometry grounded deep structured tokens model for feed-forward 3-D structure prediction.

6.2 Large Reconstruction Models (LRM)

LRM [77] trains a large transformer on objaverse, enabling single-image-to-3D in under 5 seconds. Instant3D [78] generates NeRF from a single image via a diffusion-conditioned LRM. OpenLRM [79] is the open-source counterpart. LGM [80] learns multi-view Gaussian generation for fast single-image 3-D. CRM [81] uses a convolutional reconstruction model.

6.3 Multi-View Stereo in the Neural Era

Classical MVS has been reinvigorated by deep learning. MVSNet [82] introduced cost-volume-based deep MVS. IterMVS [83] and Uni-MVSNet [84] improved accuracy and generalisation. ElasticFusion [**?**] pioneered dense SLAM with deformable surfaces. iSDF [**?**] enables online real-time SDF reconstruction. These methods provide the geometry backbone for downstream NVS.

7 Generative Paradigms for NVS

7.1 3-D-Aware GANs

GRAF [85] and pi-GAN [86] condition radiance-field generators on latent codes, producing multi-view consistent images. EG3D [88] introduced the triplane backbone and a super-resolution network, dramatically improving quality and speed. StyleNeRF [89] integrates NeRF into a StyleGAN generator. GIRAFFE [87] decomposes scenes into compositional neural feature fields. These methods established implicit 3-D supervision via GAN training on 2-D images.

7.2 Diffusion-Based Single-View and Sparse-View NVS

Pose-conditioned diffusion. Zero-1-to-3 [91] fine-tunes a latent diffusion model on multi-view data, conditioning on relative pose for single-image NVS. Zero123++ [94] improves consistency across views. One-2-3-45 [93] lifts diffusion outputs to 3-D via MVS. SyncDreamer [92] jointly denoises multiple views with shared 3-D attention. MVDiffusion [95] generates multi-view consistent images via a correspondence-aware attention mechanism. Wonder3D [96] generates cross-domain diffusion outputs (colour + normal) for high-quality 3-D reconstruction. EscherNet [97] uses positional encoding for arbitrary view generation.

Reconstruction-diffusion hybrids. Cat3D [108] generates consistent multi-view images then feeds them to a NeRF/3DGS pipeline. ReconFusion [109] integrates pixel-level diffusion priors into sparse-view NeRF. ZeroNVS [98] enables zero-shot NVS from a single in-the-wild image. Free3D [99] conditions a video diffusion model on camera trajectories for single-image novel-view sequences.

7.3 Score Distillation and Text-to-3D

DreamFusion [102] optimises a NeRF to maximise the log-likelihood under a pre-trained 2-D diffusion model. Magic3D [103] uses coarse-to-fine optimisation with mesh refinement. ProlificDreamer [104] introduces variational score distillation for higher diversity and fidelity. Fantasia3D [105] disentangles geometry and appearance for editable results. GaussianDreamer [106] uses SDS to initialise and optimise 3DGS primitives. Lucid Dreamer [107] improves SDS via interval score matching.

Table 6: Generative NVS methods.

Method	Venue	Backbone	Input	3D Cons.	3D Output	Key Idea
GRAF [85]	NeurIPS'20	NeRF-GAN	Noise	∼	NeRF	Radiance field GAN
pi-GAN [86]	CVPR'21	NeRF-GAN	Noise	∼	NeRF	Periodic implicit GAN
GIRAFFE [87]	CVPR'21	Feature F.	Noise	∼	Neural	Compositional fields
EG3D [88]	CVPR'22	Triplane	Noise	✓	Tri.	Triplane+SR
StyleNeRF [89]	ICLR'22	StyleGAN	Noise	∼	NeRF	StyleGAN+NeRF
Zero-1-to-3 [91]	ICCV'23	LDM	1 img	∼	None	Pose-cond. LDM
SyncDreamer [92]	arXiv'23	LDM	1 img	✓	None	Sync multi-view
One-2-3-45 [93]	NeurIPS'23	LDM+MVS	1 img	✓	Mesh	Diff→3D
Zero123++ [94]	arXiv'23	LDM	1 img	∼	None	Improved Zero123
MVDream [101]	ICLR'24	LDM	Text	✓	None	3D-aware MV diff.
Wonder3D [96]	CVPR'24	LDM	1 img	✓	None	Cross-domain diff.
DreamFusion [102]	ICLR'23	LDM+NeRF	Text	∼	NeRF	SDS
Magic3D [103]	CVPR'23	LDM+Mesh	Text	∼	Mesh	Coarse-to-fine SDS
ProlificDreamer [104]	NeurIPS'23	LDM+NeRF	Text	∼	NeRF	Variational SDS
GaussianDreamer [106]	CVPR'24	LDM+3DGS	Text	∼	3DGS	SDS+3DGS
Cat3D [108]	arXiv'24	MV-Diff	Sparse	✓	3DGS	MV diff+recon
ReconFusion [109]	CVPR'24	MV-Diff	Sparse	✓	NeRF	Diff prior+recon
GeNVS [100]	ICLR'23	LDM	Sparse	∼	NeRF	NeRF-cond diff.

7.4 Multi-View Diffusion Models

MVDream [101] trains a multi-view diffusion model on 3-D-consistent data for 3-D-aware generation. SyncDreamer [92] uses feature synchronisation across views. Wonder3D [96] generates colour and normal maps jointly. Consistent View Synthesis [**?**] enforces long-range temporal and viewpoint consistency via diffusion. GeNVS [100] conditions a diffusion model on a NeRF scene representation.

Table 6 summarises generative NVS methods.

8 Towards Generative World Modeling

8.1 Dynamic Scenes and 4-D World Representation

The key step from NVS to world modeling requires modelling scene dynamics. Beyond the dynamic NeRF and 4DGS methods surveyed above, recent work targets *generalizable* 4-D scene generation.

ViewCrafter [110] conditions a video diffusion model on camera trajectories, producing temporally coherent novel-view sequences. 4DiM [111] generates 4-D scenes from sparse images via a 4-D diffusion model. SV4D [112] extends stable video diffusion to 4-D object generation. Consistent4D [113] produces temporally consistent 4-D content via a cascade diffusion approach. DreamGaussian4D [114] combines SDS with 4DGS for text-driven 4-D generation.

8.2 Neural Scene Simulators for Autonomous Driving

Autonomous driving is a prime application of NVS-as-world-model.

UniSim [118] constructs a closed-loop neural sensor simulator for urban driving, rendering novel views with controllable scene edits. MARS [119] composes NeRF-based backgrounds and foreground actors for urban scene simulation. OmniRe [120] builds an all-in-one neural driving simulation framework. EmerNeRF [121] is a self-supervised approach for emergent dynamic modelling in driving scenes. Street Gaussians [122] applies 3DGS to urban driving. PVG [123] uses periodic vibration Gaussians for real-time driving scene reconstruction.

8.3 Generative World Models and Foundation Models

Large-scale world foundation models represent the apex of this trajectory.

GAIA-1 [117] casts world modeling as unsupervised sequence prediction on video, text, and action tokens for autonomous driving. Genie [124] learns to generate interactive 2-D game worlds from unsupervised video. Genie 2 [125] generates persistent 3-D environments from a single image and supports interactive first-person navigation. Cosmos [126] trains world foundation models on internet-scale video for physical world simulation. Pandora [127] interleaves language and video

Table 7: World modeling and neural scene simulation methods.

Method	Year	3D Aware	Video	Action	Domain	Key Feature
GAIA-1 [117]	2023	∼	✓	✓	Driving	Seq. token WM
UniSim [118]	2023	✓	✓	✓	Driving	Sensor sim.
MARS [119]	2023	✓	✓	∼	Driving	Compositional
Genie [124]	2024	–	✓	✓	Games	Latent action
Genie 2 [125]	2024	✓	✓	✓	Open-world	Interactive 3D
WonderWorld [129]	2024	✓	–	–	Open-world	Instant 3DGS
ViewCrafter [110]	2024	✓	✓	–	Open-world	Cam-cond video
Cosmos [126]	2025	✓	✓	✓	Physical	WFM training
Street Gaussians [122]	2024	✓	✓	∼	Driving	3DGS driving
OmniRe [120]	2024	✓	✓	✓	Driving	All-in-one

Table 8: NVS vs. World Modeling capability boundary.

Capability	NVS (static)	World Model
Viewpoint-consistent rendering (D1)	✓	✓
Sparse/single-image input (D2)	∼	✓
Temporal dynamics (D3)	–	✓
Action/agent conditioning (D4)	–	✓
Physical plausibility (D5)	–	✓
Cross-scene generalisation	∼	✓
Causal scene understanding	–	✓

tokens for interactive world generation. WonderJourney [128] generates perpetual scene explorations via text-guided chains of image generation. WonderWorld [129] generates interactive 3-D scenes with instant 3DGS from a single image.

Table 7 compares world modeling methods.

8.4 Embodied AI and Robotics

NVS provides the spatial grounding essential for embodied AI systems. Dex-NeRF [130] uses NeRF for transparent-object grasping. iNeRF [131] inverts NeRF for 6-DoF pose estimation. NeRF-Supervisor [132] uses NeRF as a view synthesis engine for robot visual planning. GaussianGrasping [133] uses 3DGS for real-time manipulation planning. RoboGSim [134] builds robot simulation environments with 3DGS.

8.5 The Boundary: NVS vs. World Modeling

Table 8 delineates the capability boundary.

9 Datasets and Benchmarks

9.1 Object-Level Datasets

ShapeNet [145]: 51K 3-D CAD models; foundation for single-image reconstruction training. **Objaverse** [146]: 800K+ 3-D objects; enables LRM-scale training. **Objaverse-XL** [147]: 10M+ 3-D assets. **Co3D** [148]: 19K+ annotated per-category real video sequences for category-level NVS. **MVImgNet** [149]: 238K multi-view real images across 238 categories.

9.2 Static Scene Datasets

NeRF-Synthetic [17]: 8 synthetic bounded objects. **LLFF** [139]: 8 forward-facing real scenes. **DTU** [140]: 124 controlled MVS scenes. **Tanks and Temples** [141]: large-scale indoor/outdoor. **Mip-NeRF 360** [20]: 9 unbounded 360° scenes. **ScanNet** [142]: 1513 indoor RGB-D sequences. **ScanNet++** [143]: DSLR + iPhone high-resolution indoor. **RealEstate10K** [144]: 10K real estate videos for generalizable NVS training.

Table 9: Key NVS datasets. RT=real, Syn=synthetic.

Dataset	Year	Type	Dyn.	GT Depth	Scale	Primary Use
ShapeNet [145]	2015	Syn	–	✓	51K obj.	Recon. training
DTU [140]	2014	RT	–	✓	124 scn.	MVS/NVS
LLFF [139]	2019	RT	–	–	8 scn.	Fwd-facing
NeRF-Synth. [17]	2020	Syn	–	✓	8 obj.	NeRF eval.
RealEstate10K [144]	2018	RT	–	–	10K vid.	Generalizable NVS
Tanks&Temples [141]	2017	RT	–	✓	Large	Reconstruction
D-NeRF [37]	2021	Syn	✓	–	6 obj.	Dynamic NeRF
HyperNeRF [40]	2021	RT	✓	–	Small	Topol. change
Mip-NeRF360 [20]	2022	RT	–	–	9 scn.	Unbounded
Neu3D [41]	2022	RT	✓	–	Human	Dyn. 3DGS
Co3D [148]	2021	RT	∼	–	19K seq.	Category NVS
ScanNet++ [143]	2023	RT	–	✓	Indoor	Generalise
Objaverse [146]	2023	Syn	∼	✓	800K obj.	LRM training
Waymo Open [153]	2020	RT	✓	✓	Urban	Driving sim.
nuScenes [154]	2020	RT	✓	∼	Urban	World model

Table 10: NVS quality on NeRF-Synthetic (Blender) and LLFF datasets. Higher PSNR/SSIM and lower LPIPS are better.

Method	NeRF-Synthetic			LLFF		
	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
NeRF [17]	31.01	0.947	0.081	26.50	0.811	0.250
Mip-NeRF [19]	33.09	0.961	0.043	26.70	0.829	0.177
Instant-NGP [23]	33.18	0.963	0.037	25.59	0.817	0.224
TensoRF [24]	33.14	0.963	0.047	26.73	0.839	0.204
Mip-NeRF360 [20]	–	–	–	29.40	0.901	0.127
3DGS [46]	33.32	0.963	0.044	27.21	0.815	0.183
Mip-Splatting [47]	34.66	0.973	0.019	27.79	0.827	0.163
Zip-NeRF [21]	–	–	–	30.38	0.921	0.100

9.3 Dynamic Scene Datasets

D-NeRF [37]: 6 monocular synthetic dynamic objects. **HyperNeRF** [40]: monocular topological changes. **NHR** [150]: multi-view human performance. **CMU Panoptic** [151]: dense multi-view social activities. **Neu3D** [41]: multi-view dynamic human sequences. **iPhone Dataset** [152]: casually-captured phone videos.

9.4 Autonomous Driving and Large-Scale

Waymo Open [153]: multi-camera + LiDAR driving. **nuScenes** [154]: 360° driving + HD maps. **KITTI-360** [155]: urban driving with panoptic labels. **Argoverse 2** [156]: high-resolution 360° driving. **PandaSet** [157]: urban LiDAR+camera dataset.

Table 9 provides a structured comparison of key datasets.

10 Evaluation Metrics and Protocols

10.1 Standard Image Quality Metrics

PSNR (peak signal-to-noise ratio): pixel-level fidelity. **SSIM** [158]: structural similarity. **LPIPS** [159]: perceptual similarity via deep features; best correlated with human judgement.

Table 10 shows representative benchmark results on NeRF-Synthetic and LLFF.

10.2 Video and World Modelling Metrics

FVD [160] (Fréchet Video Distance): distributional quality of generated video sequences. **FID** [161]: Fréchet Inception Distance for image quality. **Temporal consistency**: average cosine similarity of DINO features between consecutive frames. **Task-specific**: for driving, closed-loop collision rate, route completion, imitation gap; for robotics, task success rate.

10.3 Limitations of Current Evaluation

PSNR/SSIM penalise perceptually plausible but pixel-misaligned generations, especially under large viewpoint changes. FVD captures perceptual quality but not geometric accuracy. No existing benchmark jointly evaluates geometry, dynamics, action responsiveness, and physical plausibility. This evaluation gap is itself a major open problem.

11 Open Problems and Research Roadmap

11.1 The Geometry–Generation Trade-Off

Reconstruction methods deliver geometric precision but lack generative priors; diffusion methods provide rich priors but can violate multi-view consistency. Methods like ReconFusion, Cat3D, and ViewCrafter begin to bridge this gap, but the fundamental tension remains. Principled solutions (*e.g.*, geometry-grounded diffusion, diffusion-seeded 3DGS) are an active frontier.

11.2 Long-Horizon Temporal Coherence

Existing 4-D methods maintain coherence over a few seconds. Two promising directions: (1) *Persistent scene memory*: Gaussian maps that accumulate observations across time and preserve global consistency; (2) *World model pre-training*: leveraging internet-scale video prediction as a temporal prior that can be fine-tuned to specific domains.

11.3 Physical Grounding

Current methods learn appearance correlations, not physical laws. Encouraging directions include: PAC-NeRF [138] for continuum mechanics integration; physics-informed deformable Gaussians for material simulation; Neural SPF [**?**] for fluid simulation. The central challenge is integrating differentiable physics into the rendering pipeline in a way that scales to complex, open-world scenes.

11.4 Open-Vocabulary Semantic Integration

Methods like LERF [135] ground CLIP features in 3-D via NeRF, enabling open-vocabulary 3-D queries. LangSplat [136] brings language grounding to 3DGS. Feature3DGS [137] distils various 2-D foundation model features. Extending these to dynamic, actionable scenes is a key step toward language-grounded world models.

11.5 Evaluation and Benchmarks

We propose a five-axis world-modelling benchmark: (1) 3-D consistency under adversarial viewpoints; (2) temporal coherence over long horizons (>30 s); (3) action responsiveness in interactive settings; (4) physical plausibility via physics-engine verification; (5) open-world generalisation from a single image.

11.6 Research Roadmap

R1. Unified architectures combining explicit 3-D with diffusion priors.

R2. Internet-scale feed-forward generalisation for open-world NVS.

R3. Differentiable physics priors in 4-D world models.

R4. Language and semantic grounding for action-conditioned generation.

R5. Comprehensive world-modelling evaluation suite.

12 Conclusion

We have surveyed Novel View Synthesis through the lens of Generative World Modeling, tracing the evolution from classical image-based rendering through NeRF, 3D Gaussian Splatting, feed-forward

reconstruction, and diffusion-based synthesis toward 4-D world generation. Our unified continuous-spectrum taxonomy organises more than 160 methods across three axes (input regime, temporal scope, controllability) and makes explicit their world-modelling implications.

The field is at an inflection point. Spatial grounding from NVS and temporal/action capabilities from world modeling are actively converging, as evidenced by systems such as Genie 2, Cosmos, ViewCrafter, and the neural driving simulators. Closing the geometry–generation trade-off, achieving long-horizon coherence, and grounding predictions in physical laws are the primary challenges ahead.

Draft v0.3—ongoing work. Content will be further expanded in subsequent versions.

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